

Tonmoy Day Sarkar

Software Engineer | Machine Learning

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SUMMARY

Software Engineering graduate (Jain Deemed-to-be University, 2024) with a strong foundation in end-to-end software development and applied machine learning. Demonstrated research practice in predictive analytics, recommendation systems, and data processing, backed by an undergraduate thesis and four peer-reviewed IEEE publications. Skilled in combining full-stack software development with core machine learning techniques, including collaborative filtering, ensemble methods, geospatial clustering, and hyperparameter tuning. Seeking to build robust, scalable AI systems and data-driven software solutions.

EDUCATION

Bachelor of Technology, Computer Engineering (Software Engineering) *Jul 2020 - Nov 2024*

Jain (Deemed-to-be University), Bangalore, India | Final grade: 8.727/10 (EQF Level 6)

- Thesis: E-commerce-based Smart Recommendation System using element-by-element collaborative filtering combined with machine learning techniques.
- Relevant coursework: Machine Learning, Data Structures & Algorithms, Computer Architecture, Theory of Computation, Database Management Systems, Cloud Computing, Information Security, Agile Software Development.

Higher Secondary Certificate (Science) *2018*

Govt. Maulana Mohammad Ali College, Tangail, Bangladesh | Final grade: 4.08/5

Secondary School Certificate (Science) *2016*

Sristy College of Tangail, Tangail, Bangladesh | Final grade: 5.00/5

RESEARCH EXPERIENCE

Student Research Assistant *Sep 2023 - Jul 2024*

Jain (Deemed-to-be University), Bangalore, India | Supervisor: Mr. Karthikeyan S., Assistant Professor

- Conducted independent and supervised research resulting in four IEEE conference publications spanning recommendation systems, churn prediction, regression modeling, and geospatial clustering.
- Applied machine learning pipelines end-to-end: data preprocessing, feature engineering, model selection, hyperparameter tuning, and evaluation.

WORK EXPERIENCE

Data Analyst Intern, Brand & Visual *Dec 2025 - May 2026*

Dhaka, Bangladesh

- Performed data analysis and visualization to support business decision-making, applying statistical and analytical methods to real datasets.

PUBLICATIONS

- M. S. Rahman, T. D. Sarkar, U. T. Mitasha, M. S. Mia and S. Karthikeyan, "E-commerce-based Smart Recommendation System using element-by-element collaborative filtering following with the Machine Learning Technology," *2024 International Conference on Communication, Computing and Internet of Things (IC3IoT)*, Chennai, India, 2024, pp. 1-6, doi: 10.1109/IC3IoT60841.2024.10550386.
- T. D. Sarkar, M. S. Rahman and K. S., "Analysis of Customer Churn for Telecom Company with SMOTE-ENN and Hyperparameter Tuning Randomized-SearchCV Technique in Advanced Machine Learning Technology," *2024*

International Conference on Advancements in Power, Communication and Intelligent Systems (APCI), KANNUR, India, 2024, pp. 1-6, doi: 10.1109/APCI61480.2024.10617375.

- T. D. Sarkar, M. S. Rahman and M. Y. Emon, "Predicting Restaurant Ratings: A Comparative Study of Linear Regression and Decision Tree Regression Models in Machine Learning Technology," *2024 4th International Conference on Advancement in Electronics & Communication Engineering (AECE)*, GHAZIABAD, India, 2024, pp. 481-486, doi: 10.1109/AECE62803.2024.10911460.
- M. S. Rahman, T. D. Sarkar and M. Y. Emon, "Comprehensive Geospatial and Statistical Analysis of Restaurant Distribution Using K-means Clustering in Machine Learning Technology," *2024 4th International Conference on Advancement in Electronics & Communication Engineering (AECE)*, GHAZIABAD, India, 2024, pp. 461-466, doi: 10.1109/AECE62803.2024.10911593.

TECHNICAL SKILLS

- Programming & Machine Learning: Python, C, C++, scikit-learn (regression, clustering, ensemble methods, SMOTE-ENN), data analysis
- Web & Databases: JavaScript, Node.js, Express.js, React.js, Next.js, MongoDB, Firebase, HTML, CSS, Bootstrap
- Tools: Microsoft Word, Hadoop (data ingestion), Git, GitHub

CERTIFICATIONS & TRAINING

- Machine Learning Virtual Internship - CodSoft, IT Service
- Certificate of Reviewing - European Journal of Computer Sciences and Informatics
- Data Analysis Using Python - IBM (Cognitive Class)
- Asia AI Odyssey Challenge - Microsoft
- Moving Data into Hadoop (BD0131EN) - Big Data University
- Data Analytics and Visualization - Accenture North America (Forge)

HONOURS & AWARDS

- Study in India (G2) Scholarship - Jain (Deemed-to-be University), 2021
- The Chancellor's Scholarship - Jain (Deemed-to-be University), 2020

LANGUAGE SKILLS

Bengali (Native), English (Professional Working Proficiency)

REFERENCES

- Dr. Arvind K. S., Associate Professor, Jain (Deemed-to-be University), India | +91 9944323386 | ks.arvind@jainuniversity.ac.in
- Mr. Karthikeyan S., Assistant Professor, Jain (Deemed-to-be University), India | +91 9543277017 | s.karthikeyan@jainuniversity.ac.in